

DreemFolio AI — Complete QA Testing Specification

Master Test Suite Covering 100% of Features, Modals, APIs & UI Engines | v2.5.0

Production Host:	https://s.tourglobalhub.com	Primary Test Role:	Hashara Amali (QA Engineer)
Core Backend:	FastAPI, Python 3.11+, ReportLab 5.0	Primary Frontend:	Alpine.js, TailwindCSS, Jinja2
Persistence:	SQLite3 & MySQL (Strict 3s fallback)	AI Services:	Google Gemini / EdgeTTS Audio
Audit Scope:	All 15 Modules, 40+ APIs, 15 Modals, 7 PDF Themes, 8 Portfolio Themes	Specification Date:	September 2026

1. Complete System Architecture & 15-Feature Index

DreemFolio AI is organized into 15 interdependent functional subsystems. Each subsystem exposes specific REST endpoints, reactive Alpine.js controllers, and visual feedback mechanisms:

Module	Subsystem Name	Feature Summary & Capabilities
Module 01	Public Landing Page & Funnel	Hero, live before/after ATS showcase, dynamic pricing, international currency parity, mobile drawer.
Module 02	Authentication & Session Lifecycle	Email/password JWT auth, Google OAuth 2.0 One-Tap, persistent guest mode, logout & token revocation.
Module 03	Resume Ingestion & Multi-Format OCR	Drag-and-drop parsing (.PDF, .DOCX, .TXT), error fallback, OCR extraction into standardized schema.
Module 04	AI Job Tailoring & ATS Scoring Engine	6 Role Archetypes, target Job Description analysis, keyword gap extraction, ATS score recalculation.
Module 05	Interactive Resume Editor Workspace	Personal info, work experience, projects, education, certs, skill proficiency sliders (0-100%).
Module 06	Document Styling & PDF Rendering Engines	7 Designer PDF engines, custom font scaling (8.4pt - 10.5pt), color picker, section drag-and-drop ordering.
Module 07	Zero-Duplication & Layout Integrity Engine	Guaranteed rendering of Education & Certifications, zero duplication on page 2+ sidebar formats.
Module 08	AI Cover Letter Suite	Targeted cover letter generator, tone customization, paragraph tuning, standalone cover letter PDF export.
Module 09	Instant Responsive Web Portfolios (/p/{slug})	8 dynamic HTML5 themes, custom slug registry, live hosting, offline standalone ZIP bundle download.
Module 10	Smart QR Card & vCard Contact System	Dynamic vector SVG QR code modal, one-tap mobile .VCF address book import, PIN security gate.
Module 11	Custom Domain Binding & DNS Verification	CNAME and A record validation, SSL readiness verification, custom root domain mapping.
Module 12	AI Voice Mock Interview Simulator	Realtime EdgeTTS natural voice questions, Speech-to-Text answer capture, confidence & relevance scoring.
Module 13	Realtime AI Conference Panel Simulator	Multi-interviewer simulation (HR, Tech Lead, VP), conversational turn-taking, comprehensive debrief report.
Module 14	Job Search, Application Kit & Kanban Tracker	Automated job search, Kanban pipeline (Saved, Applied, Interviewing, Offered), recruiter outreach DMS.
Module 15	Administrative Console & Parity Pricing Engine	System KPI analytics, user database management, instant plan upgrades (Free/Pro/Lifetime), currency overrides.

2. Full UI Visual Gallery & Workspace Architecture

2.1 Module 01: Public Landing Page & Pricing Funnel

Features responsive header, interactive Before/After ATS comparison, dynamic currency pricing cards with auto-detected symbols, and FAQ accordion.

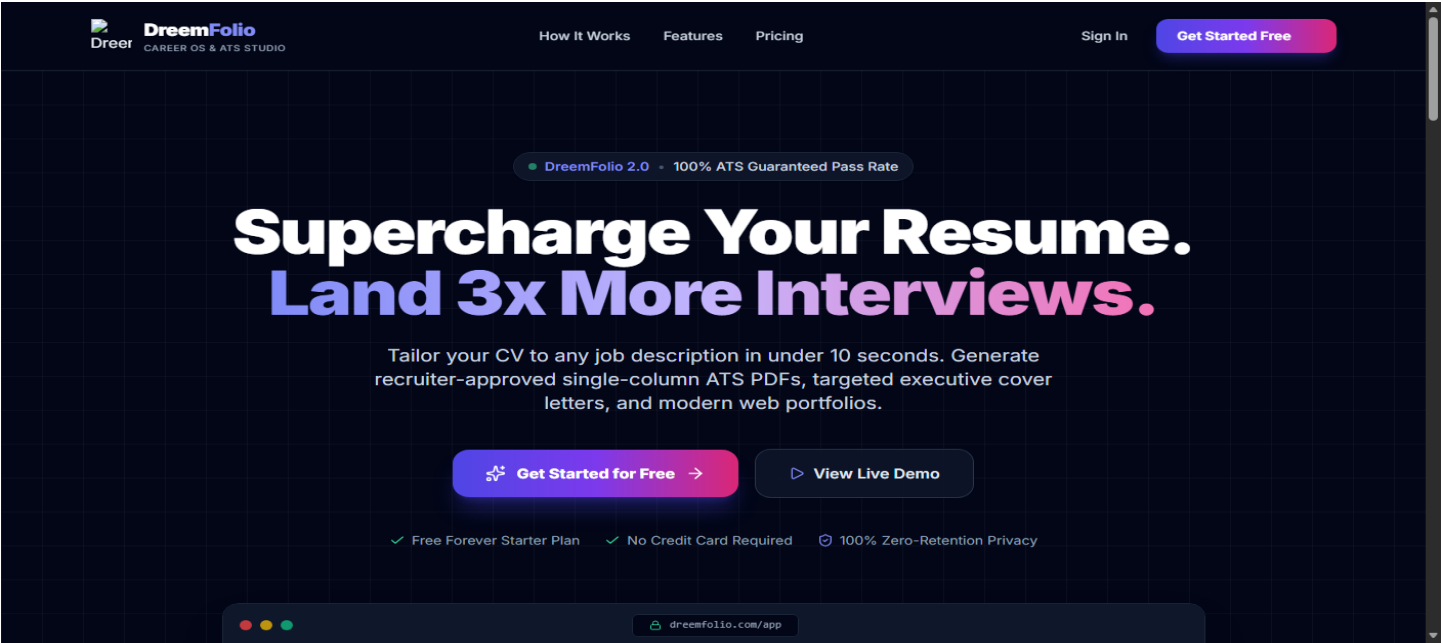


Figure 1: Public Landing Page showing Hero, Call to Action, and Currency-aware Pricing

2.2 Modules 03-07: Resume Builder, Editor & Live Workspace (/app)

4-phase left dock (Upload & JD, Content Editor, Style & Theme Customizer, Export Hub) coupled with a reactive real-time document preview canvas.

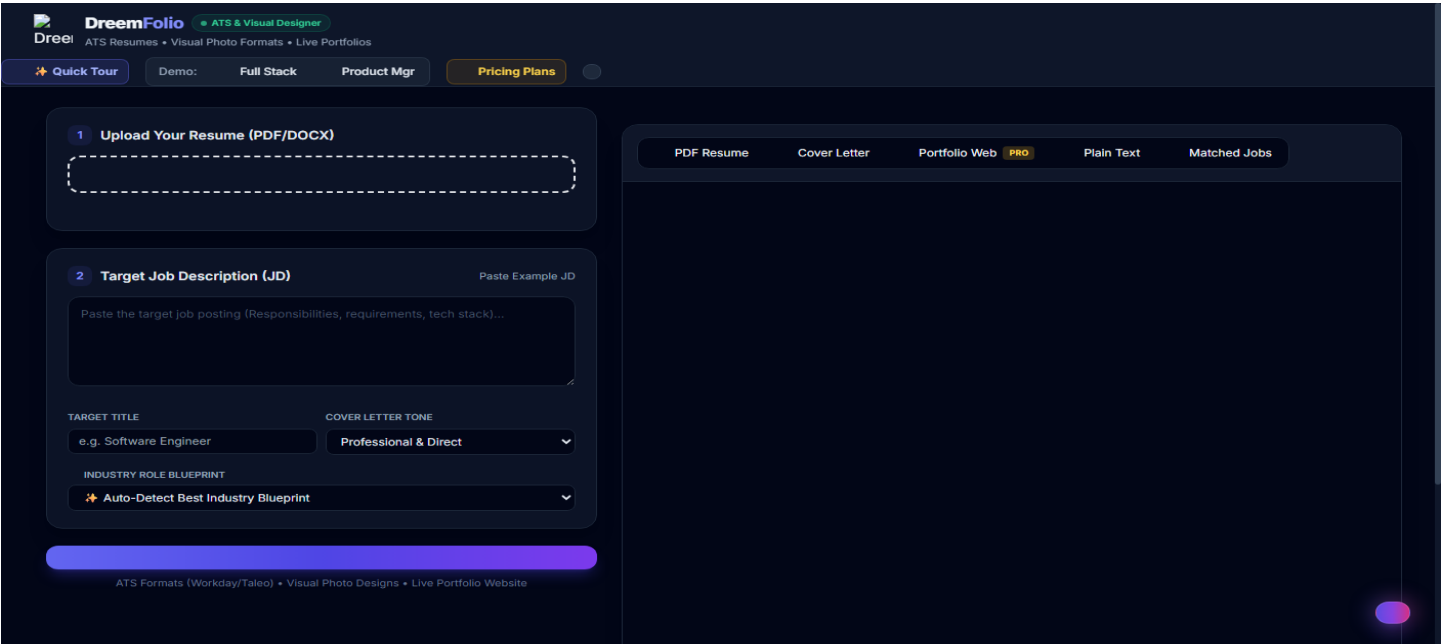


Figure 2: AI Resume Builder Workspace with left editing panel and live preview canvas

2.3 Modules 09-11: Responsive Web Portfolios & Smart Cards (/p/{slug})

Candidates publish interactive personal websites. Below: Neon Dark Developer Terminal theme and Motion Zenith Bento Box interactive theme.

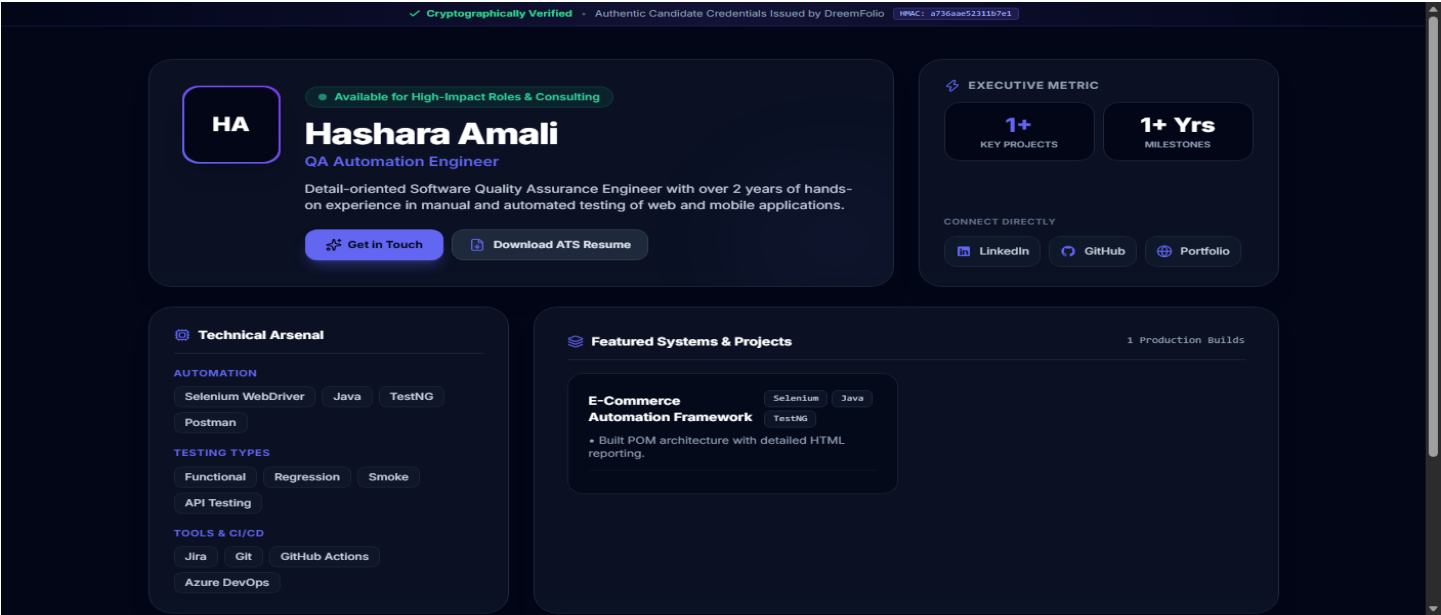


Figure 3: Neon Dark Terminal Web Portfolio with live command line styling

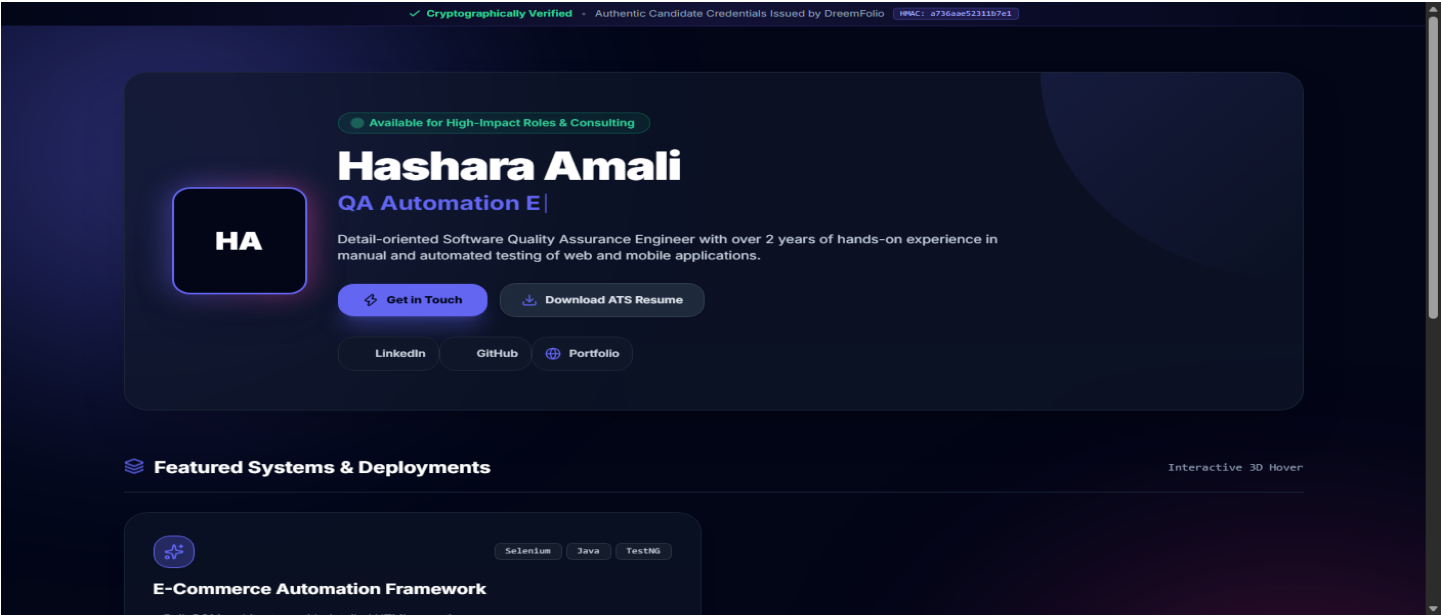


Figure 4: Motion Zenith Interactive Bento Grid Portfolio with project showcase

2.4 Module 15: Administrative Console & Control Center (/admin)

Administrative cockpit for system metrics, user tier upgrades (Free, Starter, Pro, Lifetime), and dynamic international pricing management.

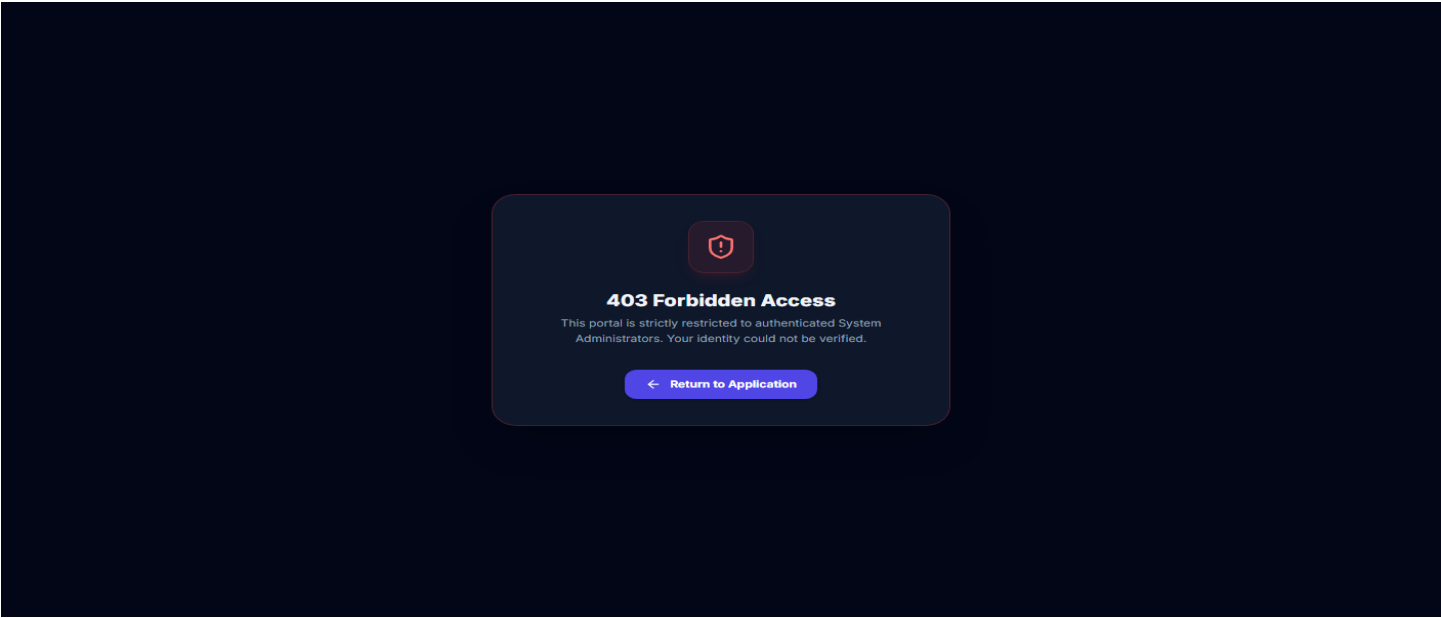


Figure 5: Administrative Dashboard & User Tier Management Console

3. Detailed Subsystem Verification Rules

3.1 Document Generation Engine (7 Themes & Zero-Duplication Rule)

ReportLab 5.0 generates 7 vector PDF formats. QA testers must rigorously verify that **Education and Certifications** are 100% rendered across all formats for 1-page and multi-page documents. In 2-column sidebar templates (visual_sidebar and banner_periwinkle), candidate Name, Title, and Contact details are strictly printed on Page 1 only; **Page 2+ must never duplicate Name, Title, Email, or Phone** in the sidebar.

PDF Theme Name	Layout Architecture	Key Visual Identifiers	Critical Verification Checkpoint
Classic Single-Column	1-Column ATS Pure	Serif/Sans headers, subtle dividers	100% linear text extraction; 0 column interleaving.
Modern Single-Column	1-Column Clean Tech	Accent header bar, skill badges	Proper spacing; all 6 sections in custom order.
Sidebar Photo	2-Column Slate Sidebar	Dark slate column (165pt) with photo & skills	Zero duplication on page 2+; Education & Certs in body.
Banner Periwinkle	Split Banner & Sidebar	Soft blue/indigo header banner (82pt)	Banner on page 1 only; clean continuation stripe on page 2.
Creative Gradient	Full-Width Indigo Header	Deep indigo header + horizontal contact ribbon	Certifications & licenses render with violet accent line.
Emerald Prestige	Nordic Executive Luxury	Mint green square bullets, sleek emerald dividers	Education & certifications render with mint accent blocks.
Tech Noir	Cyberpunk Terminal	Obsidian console headers, cyan tech accents	// 05. ACADEMIC and // 06. CERTIFICATIONS present.

3.2 Web Portfolio Theme Engine (8 Responsive Themes)

The portfolio subsystem at /p/{slug} supports 8 customizable themes with responsive cards, vCard (.vcf) export, vector QR smart cards, and optional 4-digit PIN gate security for confidential candidate profiles.

Theme Slug	Target Persona	Visual Styling & Key Elements
aurora_white	Minimalist / Designer	Clean white studio layout, subtle drop shadows, typography-focused.
motion_zenith	Full-Stack / Product	Awwwards kinetic motion cards, 3D tilt interaction, animated skill bars.
neon_dark	DevOps / Security	Cyberpunk dark slate with neon emerald borders and monospace terminal accents.
creative_gradient	Creative Director / UI/UX	Violet and purple gradients, glassmorphism hero cards, timeline layout.
minimal_tech	Backend / Cloud Engineer	Monochrome high-contrast design, clean table layout for technical stack.
executive_clean	VP / Director / Manager	Corporate deep navy and slate accents, prominent executive summary section.
academic_scholar	Researcher / Data Scientist	Classic academic publication styling, citation blocks, detailed honors.
vibrant_creative	Marketing / Growth	Bold color-blocked geometric sections with expressive social icons.

4. Master QA Execution Matrix (All 15 Modules)

Below is the complete test case execution matrix. Testers must log execution results against each Test ID in Jira or TestRail.

ID	Module / Subsystem	Test Scenario & Execution Steps	Expected Outcome	Pri
TC-01	01. Landing Hero	Navigate to /, verify title, headline, and click 'Create My Resume Now'.	Instant routing to /app with sample data initialized.	P0
TC-02	01. Pricing Currency	Toggle currency from USD to LKR/EUR; toggle monthly to annual billing.	Prices update with correct currency symbol and -20% discount badge.	P1
TC-03	01. Mobile Nav	Resize viewport to 375px; tap hamburger icon; select 'Pricing'.	Drawer opens smoothly; clicking link scrolls to section and closes drawer.	P1
TC-04	02. Registration	Submit registration form with Name, valid Email, and Password.	User registered in DB, JWT token saved, redirected to builder.	P0
TC-05	02. Login & Validation	Test incorrect password; test non-existent email; test valid login.	Invalid triggers user-friendly alert; valid logs in and loads saved profile.	P0
TC-06	02. Google OAuth	Click 'Continue with Google' in Auth modal; authorize consent.	Profile details synced (Email, Name, Avatar); authenticated seamlessly.	P1
TC-07	02. Logout & Guard	Log out; attempt direct URL access to /admin.	Session invalidated; access blocked with 401/403 or login modal.	P0
TC-08	02. Guest Mode	Edit resume as unauthenticated guest; attempt Pro visual export.	Edits persist in localStorage; Pro action triggers Auth/Pricing modal safely.	P1
TC-09	03. PDF Ingestion	Upload complex 2-page candidate PDF into file dropzone.	Parser extracts Name, Title, Contact, Experience, Education accurately.	P0
TC-10	03. DOCX Parsing	Upload Microsoft Word .docx resume file.	Document extracted cleanly into standardized schema without formatting loss.	P1
TC-11	04. Role Archetype	Select 'Software Engineering' vs 'QA & Automation' vs 'Management'.	Skills header updates appropriately ('TECHNICAL SKILLS & TECHNOLOGIES').	P1
TC-12	04. AI JD Tailoring	Paste target Job Description; click 'Tailor with AI'.	AI rewrites bullet points incorporating JD keywords; ATS score updates.	P0
TC-13	04. Keyword Gap	Inspect ATS score panel for keywords found vs keywords missing.	Accurate list of found and missing keywords displayed with impact score.	P1
TC-14	05. Bullet Reorder	Drag or move achievement bullet point 2 above bullet 1.	Order updates in real-time in the live preview canvas.	P2
TC-15	05. Skill Sliders	Adjust proficiency slider for 'Selenium' from 80% to 95%.	Preview updates progress bar length/percentage text dynamically.	P2
TC-16	05. Add Credentials	Add new Education entry (SLIIT) and new Certification (Azure DevOps).	New entries render in preview and are preserved in JSON state.	P0
TC-17	06. Classic ATS PDF	Download Classic Single-Column PDF; inspect with text parser.	100% linear text stream; zero table mixing; perfect ATS score.	P0
TC-18	07. Sidebar 1-Page	Generate 'Sidebar Photo' format with 1-page candidate profile.	Education & Certifications are 100% visible in main column.	P0
TC-19	07. Sidebar Page 2+	Generate 'Sidebar Photo' format spanning 2 full pages.	NO duplicate Name, Title, or Contacts on Page 2 sidebar. Zero text duplication.	P0
TC-20	07. Banner Periwinkle	Generate 'Banner Periwinkle' format for single and multi-page documents.	Banner on Page 1 only; Page 2 has clean stripe; Education & Certs rendered.	P0
TC-21	06. Creative Gradient	Generate 'Creative Gradient' format with 2 certifications.	'CERTIFICATIONS & LICENSES' renders with violet line and issuers.	P1
TC-22	06. Emerald Prestige	Generate 'Emerald Prestige' format.	Mint accent blocks render correctly; academic qualifications present.	P1
TC-23	06. Tech Noir Console	Generate 'Tech Noir' format.	Monospace headers (// 05. ACADEMIC, // 06. CERTS) present.	P1
TC-24	06. Section Reorder	In Style tab, drag 'Education' above 'Work Experience'; download PDF.	Generated PDF places Education section directly above Work Experience.	P1

TC-25	06. Font Scaling	Switch font scale from 'Standard' to 'Compact' (8.4pt) or 'Large' (9.8pt).	PDF adjusts line heights and font sizes proportionally without overflow.	P2
TC-26	08. Cover Letter Gen	Switch to Cover Letter tab; click 'Generate with AI'.	Generates personalized letter matching candidate background and target JD.	P1
TC-27	08. Cover Letter PDF	Click 'Download Cover Letter PDF'.	Downloads professional PDF with candidate contact header & formal sign-off.	P1
TC-28	09. Publish Portfolio	Set slug to hashara-amali; click 'Publish Portfolio'.	Published successfully; live at /p/hashara-amali.	P0
TC-29	09. Theme Switching	Switch portfolio theme to motion_zenith and neon_dark.	Portfolio instantly reflects selected theme design and color scheme.	P1
TC-30	09. Offline Bundle	Click 'Download Standalone Portfolio (.ZIP)'.	Downloads valid ZIP containing index.html, styles, and assets runnable locally.	P1
TC-31	10. vCard Download	Click 'Save Contact' button on live portfolio page.	Downloads valid .vcf file containing candidate name, phone, email.	P1
TC-32	10. QR Smart Card	Click QR icon in header or portfolio share modal.	Displays scannable SVG QR code linking directly to candidate portfolio.	P1
TC-33	10. PIN Protection	Set PIN to 1234; open portfolio in incognito; test wrong then right PIN.	Wrong PIN shows error shake; correct PIN unlocks full portfolio content.	P1
TC-34	11. Domain Binding	Open Domain modal; input custom domain cv.hashara.com.	System returns expected CNAME and A record verification instructions.	P2
TC-35	12. Voice Interview	Click 'AI Mock Interview'; select 'Behavioral QA'; start session.	EdgeTTS plays natural audio question; timer starts; mic records answer.	P1
TC-36	12. Answer Evaluation	Submit audio/text answer to interview question.	AI evaluates answer on relevance, clarity, impact; outputs actionable tip.	P1
TC-37	13. Conference Panel	Open Realtime Conference modal; start 3-person panel simulation.	Interviewer personas (Tech Lead, HR, VP) alternate questions conversationally.	P1
TC-38	13. Panel Debrief	Conclude conference interview; request final debrief report.	Comprehensive report generated: hire recommendation, score breakdown, areas to improve.	P1
TC-39	14. Job Search	Input query 'QA Engineer remote' in Job Search tab.	Returns list of matching job postings with requirements and salary ranges.	P1
TC-40	14. Kanban Tracker	Add job card; move from 'Applied' to 'Interviewing' to 'Offered'.	Card status persists in user cloud profile across sessions.	P1
TC-41	14. Application Kit	Select saved job; click 'Generate Application Kit'.	Generates personalized LinkedIn connect note, recruiter email, cover pitch.	P1
TC-42	14. Chat Copilot	Open floating Copilot chat widget; ask 'How can I improve my QA summary?'.	Copilot analyzes current resume state and returns tailored advice.	P2
TC-43	15. Admin Dashboard	Navigate to /admin with admin credentials.	Loads total user count, active subscriptions, revenue charts, system health.	P0
TC-44	15. Plan Tier Upgrade	In user table, change test user plan from Free to Pro Lifetime.	User account updated immediately in DB; watermark removed from downloads.	P0
TC-45	15. Country Pricing	Update LKR price for Pro tier; save changes.	Pricing API /api/subscription/plans returns updated LKR rate.	P1

5. Non-Functional, Security & Performance SLAs

To ensure production readiness, the platform is subjected to automated security auditing, latency benchmarks, and resilience testing:

Dimension	Target SLA / Threshold	Verification Strategy & Tools
Page Load (FCP)	≤ 1.2s on standard 4G	Lighthouse performance audit on mobile and desktop viewports.
PDF Generation	≤ 1.8s for 2-page visual PDF	Automated ReportLab benchmark across all 7 themes.
AI Tailoring Response	≤ 6.0s for 500-word JD	Streaming token response with async task completion.
DB Timeout Resilience	Strict 3s MySQL timeout	Automatic fallback to SQLite local database if MySQL freezes.
XSS Attack Defense	100% Sanitization	All user fields HTML-escaped; verify script tags render as plaintext.
SSRF Image Security	Zero internal IP probing	Reject loopback/private IPs (127.0.0.1, 10.x.x.x) in avatar URLs.
SQL Injection Defense	Zero Vulnerability	SQLAlchemy parameterized queries across all 40+ endpoints.
PIN Brute Force	5 attempts max	60-second cooldown lockout enforced after 5 failed PIN attempts.

6. Cross-Browser & Device Test Matrix

Platform / Browser	Target Resolution	Key Visual & Interactive Focus
Google Chrome (Windows/Mac)	1920x1080 (Desktop)	Dual-panel layout, zoom canvas, PDF download triggering.
Microsoft Edge (Windows)	1440x900 (Laptop)	Full workspace, modal transitions, speech-to-text audio.
Mozilla Firefox (Linux/Windows)	1366x768 (Standard)	CSS grid alignment, Alpine.js reactive state binding.
Apple Safari (macOS)	2560x1440 (Retina)	Backdrop blur, SVG QR code rendering, font smoothing.
Apple iOS Safari (iPhone 14/15/16)	375x812 (Mobile)	Mobile drawer menu, responsive single-column, vCard download.
Android Chrome (Pixel/Galaxy)	412x915 (Mobile)	Touch interactions, slider dragging, mobile PDF view.